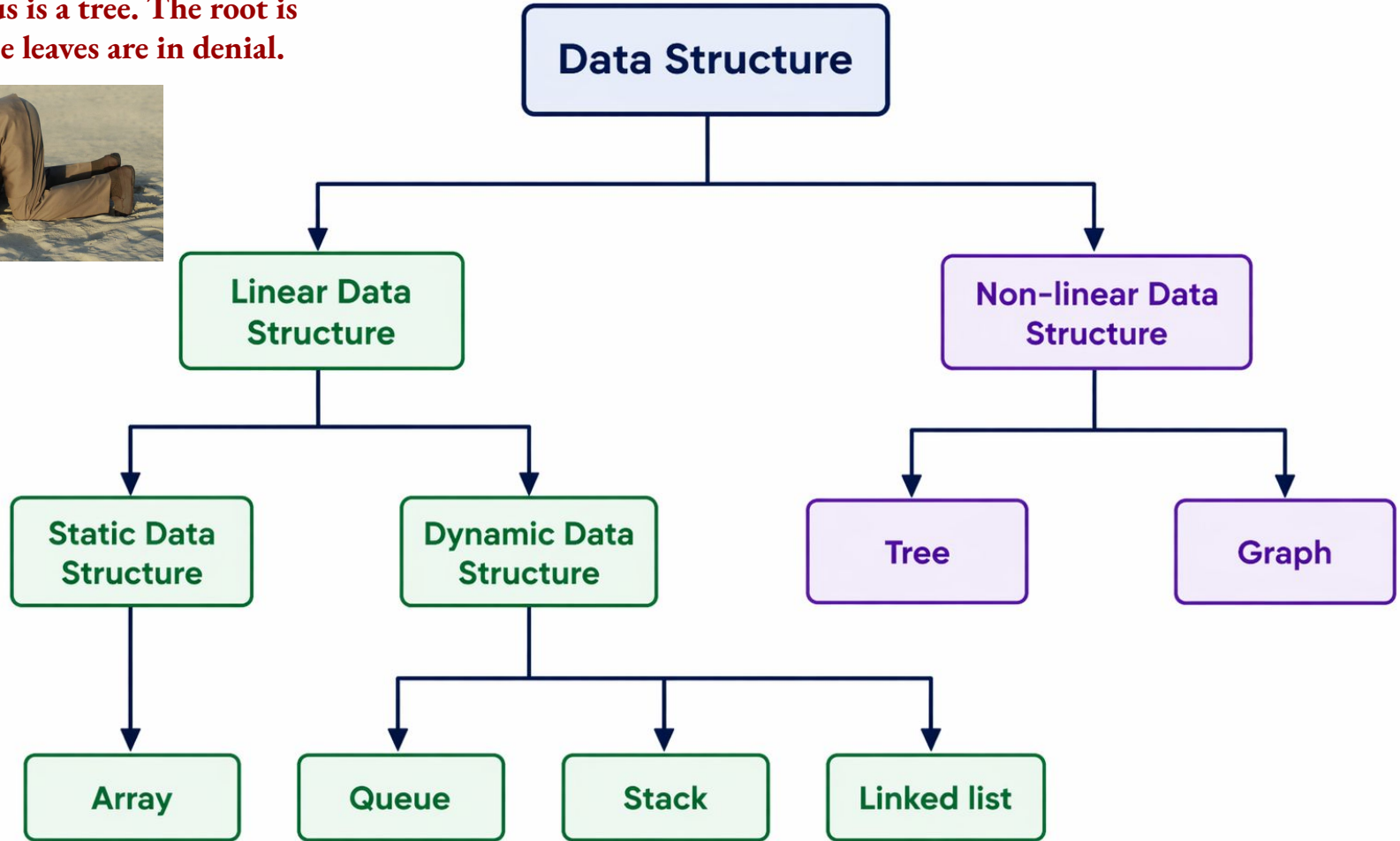
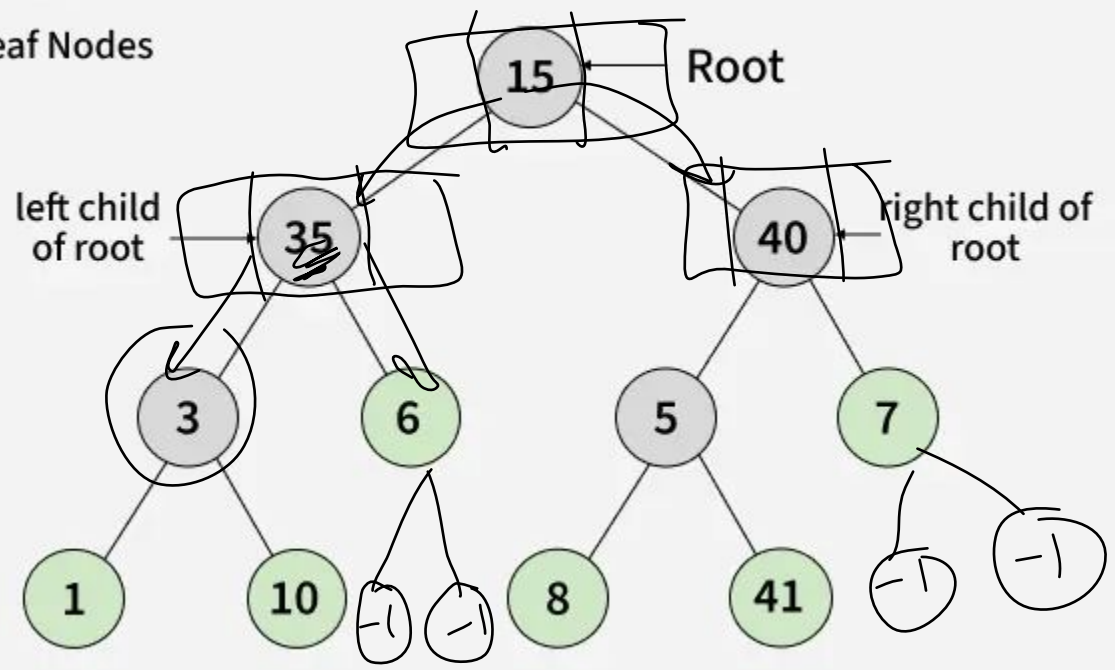


The syllabus is a tree. The root is known. The leaves are in denial.



15	35	40	3	6	5	7	1	10	-1	-1
Constructing Trees [Input and Output?]							8	41	-1	-1

 → Internal Nodes
 → Leaf Nodes



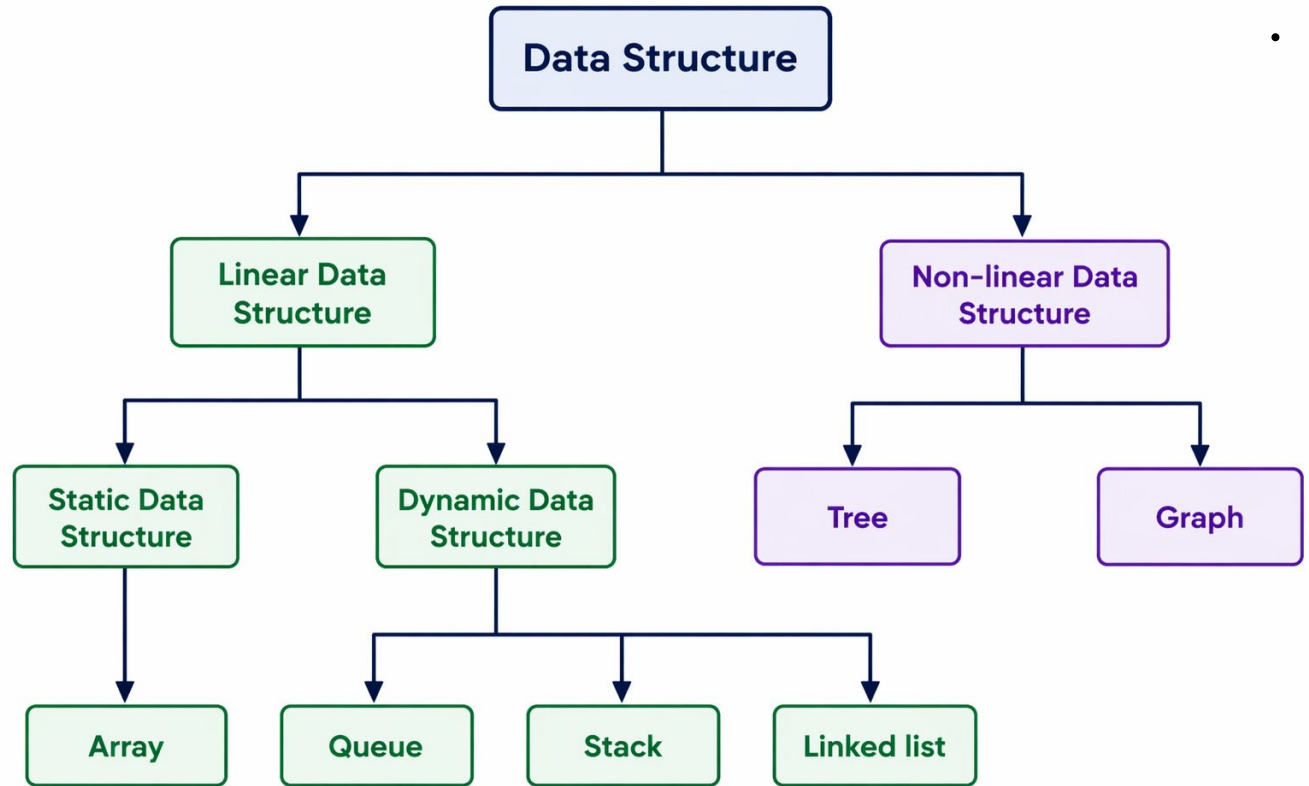
BFS BHAVANA

(BREADTH FIRST SEARCH)

Let me
cover **everything**
lightly first...
then **panic** later.

STUDY PLAN (BFS)

- ✓ See all topics
- ✓ Know a little bit of all
- ✓ Don't leave anything out
- ✓ Attempt all questions



Data Structure

Linear Data Structure

Non-linear Data Structure

Static Data Structure

Dynamic Data Structure

Tree

Graph

Array

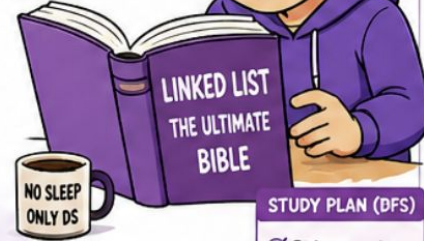
Queue

Stack

Linked list

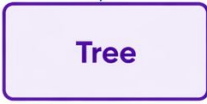
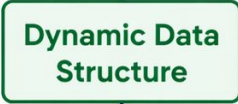
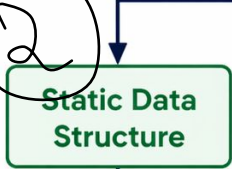
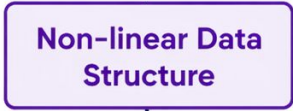
DFS DEEPAK (DEPTH FIRST SEARCH)

Let me
master one thing
completely...
others? God knows.



STUDY PLAN (DFS)

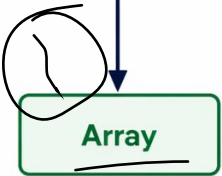
~~Static~~
~~dynamic~~
~~ds~~
~~lds~~
~~nllds~~
~~tl~~
~~g~~



{

```
process(topic)
cover(left child)
cover(right child)
```

Root
left
right

$$\text{COVER}(DS)$$


$\text{Cover}(\text{Linear}) \rightarrow C(L)$
 $\rightarrow C(R)$
 $\rightarrow \text{Process}$

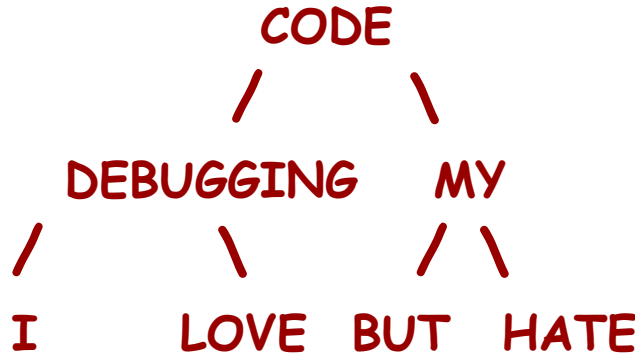
Find all traversals: Level order, Inorder, Preorder, Postorder

PO(Code)

Code

Preorder(node)
{
 print(node)
 Preorder(left)
 Preorder(right)
}

PO(Debugging)
PO(My)



Draw the binary tree whose in-order traversal is DBEAFC
and whose pre-order traversal is ABDECF

What is common in three different types of traversals (Inorder, Preorder and Postorder)?

- a) Root is visited before right subtree
- b) Left subtree is always visited before right subtree
- c) Root is visited after left subtree
- d) All of the above
- e) None of the above



<https://colab.research.google.com/drive/11qcVIJiOXmVOHAZMtc8UEfB1TjgDFYCL?usp=sharing>

```
typedef struct BST
```

```
{
```

```
}BST;
```

```
main()
```

```
{
```

```
    BST T;
```

```
    init(T);
```

```
    insert(1000,T);
```

```
    insert(9000,T);
```

```
}
```